CSCE 483 NOTEBOOK

Project: Drop-off Optimization System

Team: The Wranglers

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Notebook Entry – 01/12/2026 - 01/21/2026

Objective / Problem Statement

Today we focused on designing a system that makes the school drop-off process smooth and fast. The goal is to reduce the time it takes for each child to be picked up to around 7 minutes per kid so parents can leave quickly and traffic doesn't build up.

Brainstorming and System Design

We are building an automated drop-off system using camera-based identification and a central database. The system will verify the parent's identity and then notify staff where to bring the child.

Components & Parts List

Hardware

- **Raspberry Pi HQ Camera (12MP)** – $55 - *Adafruit Industries, ID 4561*
- **6mm Wide Angle Lens** – $50 - *Adafruit Industries, ID 4563*
- **3D printed housing** - *To protect from rain and weather*
- **Tripod**
- **Battery pack**
- **7-inch HDMI Display** – $38 - *Amazon: HAMTYSAN Raspberry Pi Screen (800x480)*

Network

- **Point-to-Point Wireless Bridge** (Ethernet-over-Radio) – $56
  To connect Raspberry Pi to the server without WiFi

Database Design

We planned a database structure to store:

- Photo ID (for facial recognition)
- Child's name
- Parent's name
- Account ID

Software System (High Level)

Website

- Built with **React/CSS/HTML/JAVA**
- **Parent Portal**
  - Parents log in and view their children
  - Shows "eligible pick-up" list
- **Admin Portal**
  - View entire database
  - Manage accounts
- **Pickup Screen**
  - Displays current pick-up status
  - Shows where to go (which station)

Facial Recognition Software

- Continuously captures photos

- Loops until a face is detected
- Runs facial recognition on detected faces

Authentication Options – Pros & Cons

Method	Pros	Cons
QR Code	- Easy to scan in rain - Controls custody issues	- Needs WiFi to generate QR code - Parents must log in every day - Slower than facial recognition
Facial Recognition	- Fastest method - Most secure - No daily work required for parents	- Hard to scan faces in rain - Most difficult to implement - Risk of inaccuracies
License Plate Recognition	- Works well in rain - No daily effort for parents	- Less secure - License plates may be hard to read - Not always visible
Pin Number	- No extra work for parents	- Slowest method - Least secure

Network Connection Design

We planned a point-to-point wireless bridge to connect the Raspberry Pi to the laptop/server without WiFi:

Diagram

Laptop ── Ethernet ── Radio)))) 50 m ((((Radio ── Ethernet ── Raspberry Pi

This allows reliable connectivity even in areas with poor WiFi.

Results / Progress

- Selected hardware components and estimated costs
- Planned database structure
- Determined software
- Developed initial workflow and process diagram
- Planned network connectivity using wireless bridge

Thoughts / Future Work

Problems / Challenges

- Facial recognition in rain may fail
- Accuracy and false positives are a major concern
- Implementation complexity is high

Next Steps

- Build a prototype using the camera + Raspberry Pi
- Test facial recognition in different weather conditions

- Evaluate QR code backup system if facial recognition fails or License plates
- Visit site

01/26/2026 – 01/28/2026: Snow Day + Career Fair

Notebook Entry - 02/02/2026

Objective / Problem Statement
Today's focus was clearly defining the **need statement** and overall **objectives** for our project. The goal was to make sure we fully understand the problem we are solving and who the system is for before moving deeper into design and implementation.

Need Statement
We identified the need for a **faster, safer, and more organized school pickup system** that reduces congestion, minimizes parent wait times, and improves student safety. Current pickup processes are inefficient, rely heavily on manual verification, and often cause long traffic backups.

Project Objectives
We established high-level objectives for the system:
- Reduce average pickup time per student
- Improve verification accuracy for authorized pickups
- Minimize traffic congestion and confusion
- Create a system that is easy for parents and staff to use
- Ensure the solution is scalable and reliable
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Requirements Brainstorming
We began identifying **engineering** and **marketing** requirements.
Engineering Requirements (Initial):
- Accurate identification of authorized parents/guardians
- Reliable communication between hardware and database
- System response time fast enough to prevent delays
- Secure storage of student and parent data
- Operate in various environmental conditions (rain, snow, lighting)

Marketing / User Requirements (Initial):
- Simple and intuitive user interface
- Minimal daily effort required from parents
- Clear instructions and feedback during pickup
- Affordable and realistic for school implementation

Results / Progress
- Completed need statement
- Defined primary project objectives
- Began drafting engineering and marketing requirements

Thoughts / Next Steps

- Refine and justify requirements
- Identify environmental and operational constraints
- Begin formal documentation of requirements

Notebook Entry - 02/04/2026

Objective / Problem Statement

Today we focused on **finalizing project requirements**, justifying them, and organizing our team's responsibilities. We also began research on existing solutions and alternative methods through literature reviews.

Requirements Finalization

We completed and justified:

- Engineering requirements
- Marketing / customer requirements
- Environmental requirements, including:
    - Operation in rain, snow, and poor lighting
    - Outdoor hardware durability
    - Network reliability in school environments

Each requirement was justified based on safety, usability, performance, and feasibility.

Project Organization

We split up the work for the project proposal, assigning sections to team members to improve efficiency and accountability. This clarified expectations and helped balance the workload moving forward.

Research & Literature Review

We began working on:

- Literature reviews related to automated pickup systems
- Research on facial recognition, QR codes, and license plate recognition
- Analysis of alternative methods and their strengths and weaknesses

This research will help support our design choices and justify why our proposed solution is effective.

Results / Progress

- Finished and justified all major requirements
- Identified environmental constraints
- Divided project proposal responsibilities
- Started literature reviews and alternative method analysis

Thoughts / Next Steps

- Continue literature review research
- Compare alternative systems more deeply

- Begin drafting proposal sections
- Start transitioning from planning to early implementation

Notebook Entry - 02/09/2026

Objective / Problem Statement

Today we focused on determining how our system will communicate and function operationally. The main goal was to select a reliable wireless method that allows fast, secure communication between the school, staff devices, and pickup identification hardware while maintaining low power usage and strong performance outdoors.

System Communication Decision (Zigbee)

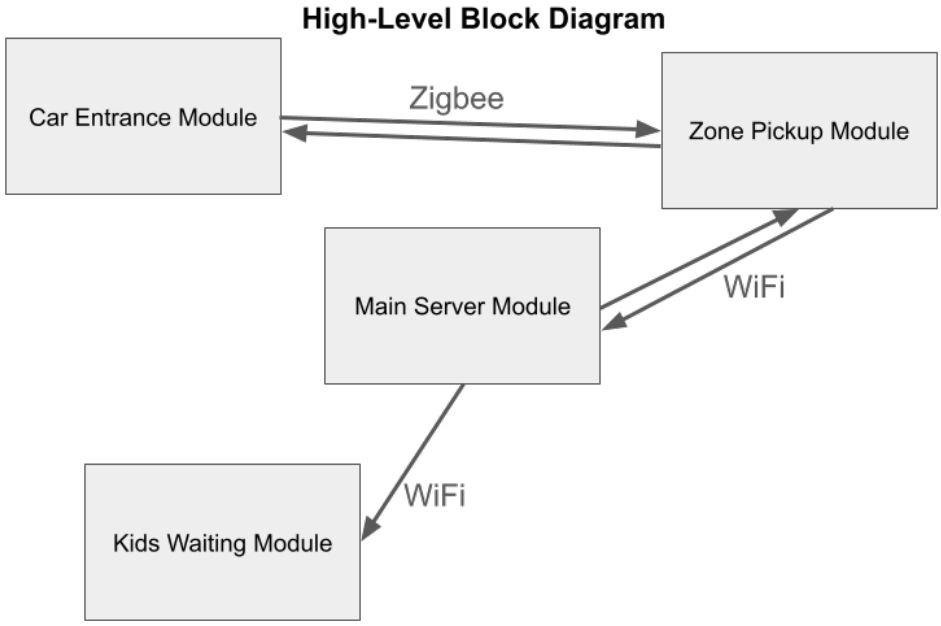
After evaluating several communication options, we selected Zigbee as the primary method for device-to-device communication within our system. Zigbee was chosen because it offers:

- Low power consumption, allowing devices to run efficiently for long periods
- Reliable short-range wireless communication suitable for school campuses
- Mesh networking capability, which allows multiple devices to relay signals and improve coverage
- Strong performance in environments with obstacles such as cars, buildings, and weather conditions
- Secure data transmission for student safety and authorization verification

This decision ensures that our system can operate consistently during busy pickup times without requiring high-bandwidth infrastructure like Wi-Fi.

System Functionality Planning

We mapped out how the system will operate during dismissal:



Proposal Presentation Development
We worked collaboratively to assemble our project proposal presentation, organizing:

- Problem definition and motivation
- Selected technology and justification
- System overview and expected operation
- Requirements identified in previous meetings
- Supporting research and comparisons to existing systems

Notebook Entry - 02/11/2026

Did our project proposal presentation and listened to other groups. Received feedback on what to include more on, for example, societal impacts, that we added to our proposal report due this week.
Began planning next steps to begin building our project, ordering items and splitting up work.

Notebook Entry - 02/16/2026

Objective / Problem Statement
Today's focus was to finalize the physical and software design so we could transition from planning into implementation. We also needed to confirm components and begin development of the user interface.

Design Finalization
We reviewed the complete system architecture, including hardware placement, communication flow, and database interaction. Final decisions were made on camera positioning, enclosure requirements, and how the Raspberry Pi will interface with the server and website.

Parts Ordering
After confirming the design, we ordered the required hardware components so assembly and testing can begin as soon as they arrive. This step marked the transition from conceptual work to physical prototyping.

Website Development Began
We started designing the structure of the web interface, including:

- Parent portal layout
- Admin dashboard concept
- Pickup display screen structure
  We focused on usability and making the interface simple and clear for school staff.

Results / Progress

- Finalized system design and configuration
- Ordered necessary hardware components

- Began front-end website design
- Established layout and functionality goals for the interface

Thoughts / Next Steps

- Receive and assemble hardware components
- Continue building the website functionality
- Begin integrating camera input with the system once parts arrive

Notebook Entry - 02/18/2026

Objective / Problem Statement

Today we focused on early testing and physical prototyping, specifically validating camera functionality and continuing development of both the interface and housing design.

Camera Testing

We tested the camera to ensure it captures clear images and can properly interface with the Raspberry Pi. Initial testing confirmed that the camera is functioning and capable of providing the image quality needed for recognition tasks.

Website Design Completion (Initial Version)

We finalized the first version of the website design, completing:

- Page layouts
- Navigation flow
- Visual structure for parent/admin interaction

  This version serves as the foundation for later backend integration.

3D Casing Development Began

We started designing the 3D-printed enclosure that will protect the hardware outdoors. Considerations included:

- Weather protection
- Camera visibility and angle
- Space for wiring and power
- Mounting stability

Results / Progress

- Successfully tested camera hardware
- Completed initial website design phase
- Began development of 3D protective casing
- Moved further into physical system implementation

Thoughts / Next Steps

- Print and refine the casing prototype
- Integrate camera with recognition software
- Begin connecting hardware to the communication network
- Start backend testing with the website interface

Notebook Entry - 02/23/2026

Objective / Problem Statement

Today we focused on advancing the user interface by creating new webpage templates and aligning the team on progress and upcoming development goals during our weekly meeting.

Web Page Template Development

Create New Account Page

We designed and implemented the template for the "Create New Account" page. Key elements included:

- Input fields for parent/guardian information
- Secure account creation workflow layout
- Clear form structure for future backend validation
- User-friendly design to support accessibility and ease of use

Kids Pickup Page

We developed the template for the "Kids Pickup" page to support safe and efficient student pickup processes. The layout includes:

- Student identification section
- Authorized pickup verification area
- Clear status indicators for pickup confirmation
- Space for future integration with camera recognition and alerts

These templates establish the structural foundation needed for backend integration and real-time system functionality.

Weekly Team Meeting

During our weekly meeting, we reviewed current progress and discussed next steps. Topics included:

- Status updates on hardware, interface, and casing development
- Coordination between frontend and backend components
- Timeline adjustments and task assignments for the upcoming week
- Planning integration milestones for camera recognition and website functionality

Results / Progress

- Created template for Create New Account page
- Developed Kids Pickup page template
- Aligned team on project progress and next steps
- Prepared interface components for backend integration

Thoughts / Next Steps

- Connect templates to backend database and validation logic
- Integrate pickup page with camera recognition system
- Conduct usability testing on account creation workflow
- Continue monitoring progress through weekly team check-ins

Notebook Entry - 02/25/2026

Objective / Problem Statement

Today we focused on improving the website's functionality and security by enhancing the volunteer page and strengthening account verification processes.

Volunteer Page Enhancements

We implemented a daily volunteer code system to improve security and ensure that only authorized volunteers can access or log their activities. Key elements included:

- Input field for entering the daily volunteer code
- Verification logic placeholder for future backend validation
- Clear instructions to guide volunteers through the process
- Layout designed for quick daily use and accessibility

This feature lays the groundwork for secure volunteer tracking and attendance verification.

Google Oath & Linked Pages Integration

We added the Google Oath verification and connected linked pages across the website to improve navigation and authentication flow. Key updates included:

- Integration placeholder for Google Oath sign-in
- Linked pages to ensure smooth navigation between sections
- Consistent header/footer structure across pages
- Improved user flow between account, volunteer, and information pages

These updates help create a more cohesive and secure user experience.

Results / Progress

- Added daily volunteer code feature to the volunteer page
- Integrated Google Oath placeholder for secure authentication
- Linked website pages for improved navigation and consistency
- Strengthened foundation for future backend security features

Thoughts / Next Steps

- Connect daily volunteer code to backend verification system
- Fully implement Google Oath authentication
- Test navigation flow across linked pages
- Conduct usability testing to ensure clear instructions and accessibility

Notebook Entry – 03/02/2026

Objective / Problem Statement

Today we focused on integrating the backend database with the **Create New Account** page to move from a static template to a fully functional account creation system. The goal was to ensure that user information submitted through the form could be stored securely and reliably in the system database.

Create Account Database Integration

We connected the Create New Account page to the backend database to enable real account creation functionality. Key elements implemented included:

- Establishing a connection between the webpage form and the database
- Configuring database queries to store new account information
- Mapping form input fields to the appropriate database columns
- Ensuring data submission occurs only when required fields are completed

This integration allows the account creation system to properly store parent or guardian information and prepares the platform for user authentication and account management features.

System Validation Preparation

While connecting the database, we also structured the system so validation checks can be implemented later. Preparations included:

- Structuring database tables for consistent data storage
- Preparing logic for future input validation
- Ensuring the system can handle multiple account entries safely
- Maintaining compatibility with future authentication features

These steps help ensure the system can scale as additional functionality is added.

Results / Progress

- Successfully connected the Create New Account page to the database
- Enabled storage of submitted account information
- Mapped webpage input fields to database columns
- Prepared system structure for future validation and authentication features

Thoughts / Next Steps

- Implement input validation to ensure accurate account data
- Add error handling for incomplete or incorrect form submissions
- Integrate authentication features for account login
- Continue testing database functionality with multiple user entries

Notebook Entry - 03/04/2026

Objective / Problem Statement

Today we focused on developing the Kids Pickup page functionality by integrating it with the database and implementing system logic for pickup stations and account validity checks. The goal was to ensure that only authorized accounts could interact with the pickup system and that pickup locations could be managed efficiently.

Kids Pickup Page Database Integration

We connected the Kids Pickup page to the backend database so that pickup requests and user information can be verified in real time. Key features implemented included:

- Database queries to retrieve account information
- Verification of account validity before allowing pickup actions
- Linking pickup actions to registered parent or guardian accounts
- Ensuring secure handling of student identification data

This allows the pickup system to confirm that a valid account is associated with each pickup request.

Pickup Stations Implementation

We added logic to support pickup stations, which organize and manage where students are collected. Key elements included:

- Station identifiers to represent different pickup locations
- Database fields to associate students with pickup stations
- Structured logic to help track pickup activity by station
- Layout support for displaying station information in the interface

This structure helps streamline the pickup process and prepares the system for future automation features.

Account Validity Checks

To improve system security, we implemented checks to confirm that accounts are valid before allowing interactions with the pickup system. This included:

- Verifying that account records exist in the database
- Ensuring account information is active and correctly stored
- Preventing unauthorized access to pick up actions
- Preparing the system for integration with authentication tools

These checks help maintain system integrity and ensure only authorized users interact with student pickup features.

Results / Progress

- Connected Kids Pickup page to the database
- Implemented account validity verification logic
- Added structure for pickup stations
- Enabled secure interaction between user accounts and pickup system

Notebook Entry - 03/16/2026

Objective / Problem Statement

Today we focused on presenting our Critical Design Review (CDR) to demonstrate the progress of our system, validate our design decisions, and receive feedback from instructors and peers.

CDR Presentation

We presented the full system design, including:

- Overall system architecture (hardware + software integration)
- Database structure and communication flow
- Website functionality and user interface design
- Pickup process workflow and station organization
- Justification for chosen technologies (Zigbee, camera system, database)

The presentation emphasized how our system improves efficiency, reduces pickup time, and ensures secure verification of authorized users.

Feedback Received

We received feedback on:

- Clarifying system scalability and real-world deployment
- Expanding discussion on reliability in outdoor conditions
- Providing more detail on error handling and fallback systems
- Improving explanation of communication between components

Results / Progress

- Successfully completed CDR presentation
- Demonstrated full system design and integration plan
- Identified areas needing improvement based on feedback

Thoughts / Next Steps

- Refine system documentation based on feedback
- Improve explanation of system reliability and scalability
- Begin implementing missing components identified during review

Notebook Entry - 03/18/2026

Objective / Problem Statement

Today we focused on improving and finalizing the Critical Design Review (CDR) report while also continuing work on system communication using Zigbee.

CDR Report Development

We worked on refining the written report to ensure it clearly communicates:

- System design and architecture
- Engineering and marketing requirements
- Justification for selected technologies
- Implementation plan and expected performance

We improved clarity, organization, and technical explanations to align with feedback received during the presentation.

Zigbee Communication Investigation
We continued exploring how Zigbee will be used in our system by:

- Understanding device pairing and communication setup
- Researching how Zigbee modules can transmit pickup data
- Evaluating how Zigbee integrates with the Raspberry Pi and server
- Identifying potential limitations such as range and interference

This step is critical to ensuring reliable communication between pickup stations and system components.

Results / Progress

- Improved and refined CDR report
- Gained deeper understanding of Zigbee communication
- Identified implementation considerations for wireless networking

Thoughts / Next Steps

- Begin testing Zigbee modules for communication reliability
- Finalize report submission
- Continue transitioning into implementation phase

Notebook Entry - 03/23/2026

Objective / Problem Statement
Today we focused on team coordination and gathering real-world insights to improve the practicality of our system design.

Weekly Team Meeting
During our weekly meeting, we:

- Set clear agendas for upcoming development tasks
- Reviewed progress across hardware, software, and UI components
- Assigned responsibilities for continued implementation
- Discussed integration timelines and testing milestones

This helped ensure alignment across the team and improved project organization.

Meeting with School Coach (Pebble Creek)
We set a meeting with the coach responsible for managing pickup operations at Pebble Creek Elementary. Questions to be asked on:

- Current pickup workflow and bottlenecks
- Common issues such as traffic congestion and delays
- Safety concerns and verification challenges
- Opportunities for improving efficiency through automation

Results / Progress

- Established clear team direction and task assignments
- Gained real-world understanding of pickup system challenges
- Identified improvements to incorporate into system design

Thoughts / Next Steps

- Adjust system workflow based on real-world feedback
- Improve usability and clarity of pickup interface
- Begin implementing features aligned with actual school needs

Notebook Entry - 03/25/2026

Objective / Problem Statement
Today we focused on improving the usability of our system by ensuring the website is fully compatible with mobile devices.

Mobile Compatibility Implementation
We updated the website design to support responsive layouts across different screen sizes. Improvements included:

- Converting multi-column layouts into single-column layouts for phones
- Adjusting button sizes and spacing for touch interaction
- Ensuring input fields are easy to access and read on smaller screens
- Improving navigation flow for mobile users

User Experience Enhancements
We prioritized accessibility and usability by:

- Increasing font sizes for readability
- Ensuring consistent layout across all pages
- Making buttons full-width for easier interaction
- Reducing clutter to improve clarity

These updates ensure parents and staff can easily use the system on mobile devices during pickup.

Results / Progress

- Successfully implemented responsive design across the website
- Improved usability for mobile users
- Enhanced accessibility and interaction design

Thoughts / Next Steps

- Test mobile performance on different devices
- Fix any layout inconsistencies or bugs
- Continue integrating backend functionality with updated UI

Notebook Entry - 03/30/2026

Objective / Problem Statement
Today we focused on setting up the backend server and establishing communication between the website and the system database.

Server Setup
We began configuring the server to handle:

- Data storage and retrieval
- Communication between frontend and backend
- Processing of user requests (account creation, pickup actions)

This involved setting up the development environment and preparing database connections.

Website Integration
We worked on connecting the website to the server by:
- Linking frontend forms to backend APIs
- Ensuring data from the website is correctly sent and received
- Testing communication between client and server
- Preparing for real-time updates in the pickup system

This step is essential for enabling dynamic system functionality.

Results / Progress
- Initiated server setup and configuration
- Began connecting website to backend system
- Established foundation for full system integration

Thoughts / Next Steps
- Complete backend API development
- Test data flow between frontend and database
- Begin integrating real-time updates for pickup system

Notebook Entry – 04/01/2026

Objective / Problem Statement
Today we focused on validating our system design through real-world feedback and continuing progress on physical prototyping.

Meeting with Coach
We met with the school coach to better understand how our system can improve efficiency. Key insights included:
- How students are currently organized during pickup
- Timing constraints and peak congestion periods
- Staff coordination challenges
- Opportunities for automation to reduce confusion

This helped us refine our system workflow to better align with actual school operations.

3D Housing Development
We continued progress on the physical prototype by:
- Finalizing the design of the 3D-printed enclosure
- Beginning the printing process
- Ensuring proper spacing for camera, wiring, and components
- Designing for durability and weather resistance

This is an important step toward building a functional outdoor system.

Results / Progress
- Gathered valuable real-world feedback for system improvement
- Began physical production of hardware enclosure
- Refined system workflow based on practical constraints

Thoughts / Next Steps

- Test 3D casing once printed
- Integrate camera into enclosure
- Continue aligning system design with real-world use

Notebook Entry – 04/06/2026

Objective / Problem Statement

Today we focused on establishing reliable communication between the server and system components while continuing development of the physical prototype.

Server Communication & Housing Completion

We successfully configured the server to communicate with the system modules, enabling data transfer between the frontend interface and backend database. This marked a key step toward full system integration.

In parallel, we finalized the design of the 3D-printed housing and sent the enclosure for printing. The housing is designed to protect the hardware in outdoor conditions while maintaining proper camera positioning and accessibility.

Results / Progress

- Established communication between server and system components
- Finalized and sent 3D housing design for printing
- Continued progress toward full system integration

Thoughts / Next Steps

- Test communication reliability across devices
- Receive and evaluate printed housing
- Begin integrating hardware into enclosure

Notebook Entry – 04/08/2026

Objective / Problem Statement

Today we focused on improving system reliability and ensuring consistent functionality across multiple devices.

Cross-Device Functionality & System Design Decision

We configured the server to function across all devices, ensuring compatibility and stable communication regardless of platform.

After evaluating system performance and reliability, we made the decision to transition to a QR code–based system. This approach simplifies implementation and improves reliability compared to previous identification methods.

Results / Progress

- Enabled server functionality across multiple devices

- Selected QR code system for improved reliability and ease of use
- Simplified overall system design

Thoughts / Next Steps

- Implement QR code generation and scanning
- Integrate QR system with database and UI
- Test system under real-world conditions

Notebook Entry – 04/13/2026

Objective / Problem Statement

Today we focused on improving the user interface and refining the physical prototype.

UI Integration & Housing Redesign

We updated the user interface to ensure full functionality and successfully connected it to the backend server. This allowed real-time interaction between the interface and system data. Additionally, we remodeled the 3D housing to improve fit, durability, and usability based on earlier design feedback.

Results / Progress

- Improved and fully functional UI connected to server
- Refined 3D housing design
- Strengthened system integration

Thoughts / Next Steps

- Test UI responsiveness and usability
- Print updated housing design
- Continue full system testing

Notebook Entry – 04/15/2026

Objective / Problem Statement

Today we focused on completing system integration and finalizing communication between all modules.

Display Setup & Full System Connection

We developed the display screen output page to show pickup information clearly and in real time.

We also completed full communication between all system modules. As part of this process, we transitioned from the previously planned communication method to WiFi to simplify connectivity and improve reliability.

Results / Progress

- Implemented display output screen

- Established full connection between all modules
- Transitioned system communication to WiFi

Thoughts / Next Steps

- Test full system workflow end-to-end
- Optimize performance and response time
- Prepare for final assembly

Notebook Entry – 04/20/2026

Objective / Problem Statement

Today we focused on completing the system and ensuring all components work together seamlessly.

Final Integration & Automation

We successfully implemented full QR code automation, allowing the system to generate, scan, and process pickup requests efficiently.
Final assembly of the project was completed, integrating hardware, software, and interface components into a fully functioning system.

Results / Progress

- Completed QR code automation
- Finalized full system assembly
- Achieved complete system functionality

Thoughts / Next Steps

- Conduct full system testing and validation
- Prepare demonstration materials
- Finalize presentation

Notebook Entry – 04/22/2026

Objective / Problem Statement

Today we focused on demonstrating the completed system and evaluating its performance.

Full System Demonstration

We conducted a full demo of the system, showcasing all components including QR code scanning, database interaction, and display output. The system successfully demonstrated its ability to streamline the pickup process.

Results / Progress

- Successfully completed full system demo
- Verified functionality of all integrated components
- Demonstrated system workflow end-to-end

Thoughts / Next Steps

- Address any minor issues identified during demo
- Finalize report and presentation materials

Notebook Entry – 04/27/2026

Objective / Problem Statement
Today we focused on finalizing all deliverables for project submission.

Report & Presentation Development
We worked on completing the final report and presentation, ensuring all sections clearly explain the system design, implementation, and results.

Results / Progress

- Continued development of final report
- Refined presentation content and structure
- Prepared for final submission

Thoughts / Next Steps

- Finalize all documentation
- Rehearse presentation

Notebook Entry – 04/29/2026

Objective / Problem Statement
Today we completed the final presentation of our project.

Final Presentation
We presented our completed system, demonstrating its functionality, design process, and impact on improving school pickup efficiency.

Results / Progress

- Successfully delivered final presentation
- Demonstrated full system capabilities
- Completed project requirements